ALL INDIA TEST SERIES

Dropper JEE Advanced

AJPA11.2/10

Test- 10

DURATION : 180 Minutes

DATE : 16/03/2025

M.MARKS : 198

ANSWER KEY

PHYSICS

1. (8)
2. (3)
3. (8)
4. (7)
5. (4)
6. (3)
7. (A, B, C, D)
8. (C, D)
9. (B, C, D)
10. (B, D)
11. (B, C)
12. (B, C)
13. (6.00)
14. (1.60)
15. (4.00)
16. (1.50)
17. (2.00)
18. (2.00)

CHEMISTRY

19. (6)
20. (2)
21. (6)
22. (3)
23. (3)
24. (5)
25. (A, B, C)
26. (A, B, C)
27. (A, C, D)
28. (A, B, C, D)
29. (B, C)
30. (B, C, D)
31. (19.00)
32. (1.50)
33. (6.00)
34. (22.40)
35. (1.50)
36. (2.40)

MATHEMATICS

37. (4)
38. (7)
39. (0)
40. (1)
41. (2)
42. (0)
43. (A, C)
44. (B, D)
45. (C, D)
46. (A, C, D)
47. (A, D)
48. (A, C)
49. (6.00)
50. (0.50)
51. (18.00)
52. (1.00)
53. (0.80)
54. (0.20)

PART-I (PHYSICS)

1. (8)

$$U = -\frac{km}{r}$$

$$K_i + U_i = K_f + U_f$$

$$\frac{1}{2}mv^2 - \frac{Km}{\left(\frac{R}{2}\right)} = 0 - \frac{Km}{3R/2}$$

$$\frac{mv^2}{2} = \frac{2Km}{R} - \frac{2Km}{3R}$$

$$\frac{mv^2}{2} = \frac{4Km}{3R}$$

$$v = \sqrt{\frac{8K}{3R}},\ v = 8 \text{ m/s}$$

2. (3)

From conservation of angular momentum

$$m\frac{1}{2}\sqrt{\frac{GM}{R}}d = mvR$$

$$\Rightarrow \sqrt{\frac{GM}{R}}d = 2vR$$

From conservation of energy

$$\frac{1}{2}m\left(\frac{1}{2}\sqrt{\frac{GM}{R}}\right)^2 = -\frac{GMm}{R} + \frac{1}{2}mv^2$$

$$\Rightarrow \frac{9GM}{4R} = v^2 = \frac{GM}{R}\frac{d^2}{4R^2}$$

$$\Rightarrow \frac{d}{R} = 3$$

3. (8)

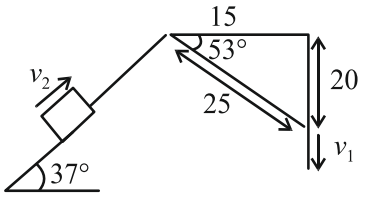
$U \propto r^2$

$F = mb^2r$

$$\frac{mv^2}{r} = mb^2r \Rightarrow v \propto r$$

$$mvr = \frac{nh}{2\pi} \Rightarrow k = 8$$

4. (7)



$$-10 \times \frac{10 \times 3}{5} + 10 \times 20 = \frac{1}{2} \times 1v_1^2 + \frac{1}{2} \times 1 \times v_2^2$$

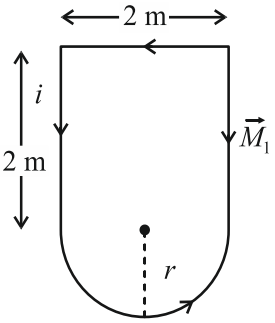
$$140 = \frac{v_1^2 + v_2^2}{2}$$

$$\frac{4v_1}{5} = v_2 \left(\text{constrain}\right)$$

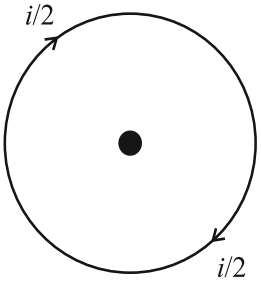
$$v_1 = \sqrt{\frac{280 \times 25}{41}}$$

$$\Rightarrow \quad v_1 = \sqrt{\frac{7000}{41}} \text{ m/sec}$$

5. (4)



$$\vec{M}_1 = \left(2^2 + \frac{\pi 1^2}{2}\right) \times 1\,\hat{k}$$



$$\vec{M}_2 = \left(\pi r^2\right)\frac{i}{2}\left(-\hat{k}\right) = -\frac{\pi}{2}\hat{k}$$

$$\vec{M} = \vec{M}_1 + \vec{M}_2$$

$$\vec{M} = \left(4 + \frac{\pi}{2} - \frac{\pi}{2}\right)\hat{k}$$

$$\left|\vec{M}\right| = 4 \text{ Am}^2$$

6. (3)

$$u = \frac{j}{m}$$

$$\varepsilon = Blv$$

$$a = \frac{F}{m} = \frac{Bil}{m} = \frac{Bl}{m} \times \frac{Blv}{R}$$

$$V_L = \frac{Ldi}{dt}$$

$$\int_0^x Bl\frac{dx}{dt} = \int_0^i \frac{Ldi}{dt}$$

$$Blx = Li$$

$$\int_u^{u/2} vdv = -\frac{Bl}{m}\int idx$$

$$= \frac{-B\ell}{m}\int_0^d \frac{Blx}{L}dx$$

$$\Rightarrow \quad d = \frac{3v^2mL}{4B^2l^2} = \frac{3j^2L}{4B^2l^2m}\left(v = \frac{j}{m}\right)$$

7. (A, B, C, D)

Let the distance of the *CM* from *A* be *x* and the *MI* about *CM* axis perpendicular be *I*. CASE I: Torque about *A*:

$$mgx\left(\frac{4}{5}\right) = \left(I + mx^2\right)\frac{g}{l} \qquad \rightarrow(1)$$

CASE II: Torque about B:

$$mg(l-x)\left(\frac{4}{5}\right) = \left(I + m(l-x)^2\right)\frac{2g}{l} \qquad \rightarrow(2)$$

Solving (1) and (2) : $x = \frac{3l}{4}$ *and* $I = \frac{3}{80}ml^2$

CASE I: $a_{cm} = \left(\frac{g}{l}\right)x = \left(\frac{g}{l}\right)\left(\frac{3l}{4}\right) = \frac{3g}{4}$

CASE II: $a_{cm} = \left(\frac{2g}{l}\right)(l-x) = \left(\frac{2g}{l}\right)\left(\frac{l}{4}\right) = \frac{g}{2}$

8. (C, D)

The acceleration of segments of rope will be downwards *a* and *a* towards the end of the tube for inside the tube. The acceleration of remaining portion of ropes will be zero. Since inelastic collisions are happening at the ends of the rope heat will be generated. Applying newton's second law on the segments of rope inside tube and the tube we get $a = \frac{g}{2}$

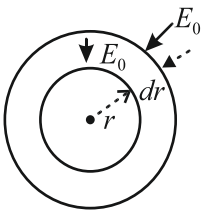
9. (B, C, D)

$$J = \sigma E = \frac{1}{\rho}E \quad J = \left(a + bh^2\right)E_0$$

Hence option (A) is incorrect. Current density increases with increases in height.

As field is directed downward electrons will flow upward (radially outward).

Now, to calculate charge density (ρ) we use gauss law (Assume a thin spherical shell as gaussian surface)



$$\phi = \phi_{1N} + \phi_{out} = \frac{q}{\varepsilon_0}$$

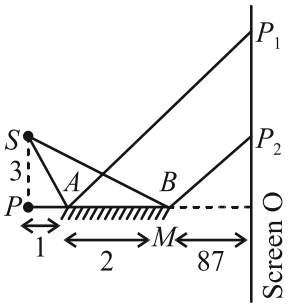
$$\Rightarrow E_0\left(4\pi r^2\right) - E_0 4\pi(r+dr)^2$$

$$= \frac{\rho\left(4\pi r^2 dr\right)}{\varepsilon_0}$$

$$\Rightarrow \rho \approx -\frac{2\varepsilon_0 E_0}{r} = -\frac{2\varepsilon_0 E_0}{R+h}$$

10. (B, D)

$$\beta = \frac{D\lambda}{d} = \frac{90\times10^{-3}\times\lambda}{6\times10^{-3}} \Rightarrow \lambda = \frac{6\times10^{-4}}{90} = \frac{60}{9}\mu m$$

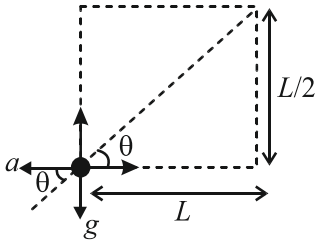


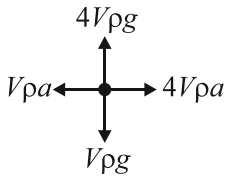
$OP_1 = 89 \times 3 = 267$ mm $\qquad OP_2 = 87$ mm

$P_1P_2 = 180$ mm

$$N = \frac{180}{0.1} = 1800$$

11. (B, C)





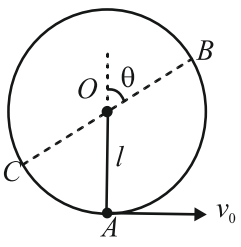
In frame of container,

$$\tan\theta = \frac{1}{2} = \frac{g}{a}$$

$\Rightarrow a = 2g$

Now, $\frac{L}{2} = \frac{1}{2}\cdot 3g\cdot t^2 \Rightarrow t = \sqrt{\frac{L}{3g}}$

12. (B, C)



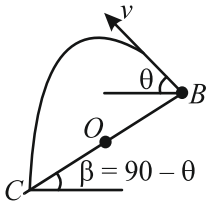
Let v = speed at B.

Tension in the string at $B = 0$

$$\therefore mg\cos\theta = \frac{mv^2}{l}$$

$$\Rightarrow v^2 = gl\cos\theta \quad \ldots\ldots\ldots\ldots\ldots\ldots (a)$$

After B, the motion of particle is that a projectile. One can consider its projectile thrown down the incline BOC. The particle hits the incline at C.



$\therefore$ Range along the incline = $BO + OC = 2 \times l$

Incline angle $\beta = 90 - \theta$

For such a projectile, you can prove that the range along the incline will be

$$\frac{2gl\cos\theta\cdot(\cos\theta)}{g\sin^2\theta} = 2l\cot^2\theta$$

$\therefore\ 2l = 2l\cot^2\theta$

$\therefore\ \cot^2\theta = 1$

Or, $\theta = 45°$

Now we can use conservation of energy between points A and B, which goes

$$\frac{1}{2}mv_0^2 = \frac{1}{2}mv^2 + mg(1+\cos\theta)$$

$$\Rightarrow v_0^2 = v^2 + 2gl\,(1+\cos\theta)$$

using (a)

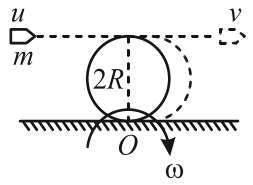
$$v_0^2 = 3gl\cos\theta + 2gl$$

$$= gl\left[3\times\frac{1}{\sqrt{2}} + 2\right]\left[\because \theta = 45^\circ\right]$$

$$= \frac{(4+3\sqrt{2})}{2}gl$$

13. (6.00)

We can assume the rolling of cylinder as pure rotation about an axis passing through 'O' (see figure). Using conservation of angular momentum about O



$(m)\,(u)\,(2R) = (m)\,(v)\,(2R) + I_0\omega \qquad \ldots.(i)$

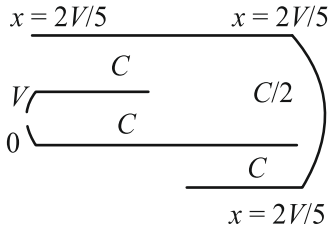
Here, $v = (2R)\,\omega$ and

$$I_0 = \frac{MR^2}{2} + MR^2 = \frac{3}{2}MR^2$$

Substituting these in Eq (i), we get

$$\omega = \frac{4mu}{(8m+3M)R} \text{ and } v = \frac{8mu}{(8m+3M)}$$

14. (1.60)



$$C(x-V) + \frac{C}{2}(x-0) + C(x-0) = 0$$

$$\frac{5x}{2} = V$$

$$x = \frac{2V}{5}$$

Q on plate connected to $+ve$ terminal

$$= C\left(V - \frac{2V}{5}\right) + C(V-0) = C_{eq}V$$

$$= \frac{8CV}{5} = C_{eq}V \quad \Rightarrow C_{eq} = \frac{1.6\varepsilon_0 La}{d}$$

15. (4.00)

$$K_{e\max} = h\nu - \phi = \frac{6.6\times10^{-34}\times7.27\times10^{14}}{1.6\times10^{-19}} - 1$$

(inner sphere)

= 2 eV

Maximum potential difference is 2 V

$$U = \frac{1}{2}CV^2 = 4\ \mu\text{J}$$

16. (1.50)

$$\mu mgv = -\left(\frac{dm}{dt}\right)L$$

$$\int_{m_0}^{m}\frac{dm}{m} = -\frac{\mu g}{L}\int_0^t(v_0 - \mu gt)\,dt$$

$$m = m_0 e^{-\frac{\mu g}{L}\left(v_0 t - \frac{1}{2}\mu g t^2\right)}$$

17. (2.00)

Let $m \propto c^x G^y h^z$ or $m = Kc^x G^y h^z$

By substituting the dimension of each quantity in both sides

$$\left[M^1L^0T^0\right]=K\left[LT^{-1}\right]^x\left[M^{-1}L^3T^{-2}\right]^y\left[ML^2T^{-1}\right]^z$$

$$\left[M^1L^0T^0\right]=\left[M^{-y+z}L^{x+3y+2z}T^{-x-2y-z}\right]$$

By equating the power of M, L and T in both sides:

$-y+z=1, x+3y+2z=0, -x-2y-z=0$

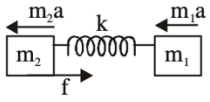
By solving above three equations, we get

$x=\frac{1}{2}, y=-\frac{1}{2}$ and $z=\frac{1}{2}$

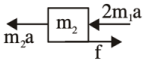
So, $m \propto c^{1/2}G^{-1/2}h^{1/2}$

18. (2.00)

When viewed from frame of plank



[initial condition]



[at the time of maximum compression]

$\because$ $2m_1a+m_2a$ = 8N and $f_{max} = \mu N$ = 10 N

$\Rightarrow$ Friction is static

$kA = m_1a \Rightarrow A = \frac{m_1a}{k} = 2$ cm

PART-II (CHEMISTRY)

19. (6)

Molarity of 1st solution = $\frac{0.2}{\frac{0.2M+18\times0.8}{d_1}}\times1000$

Molarity of 2nd solution

$=\frac{0.1}{\frac{0.1M+0.9\times18}{d_2}}\times1000$

$\Rightarrow$ $d_2 = 0.8d_1$

$\frac{1}{2}\times\frac{0.2\times1000}{0.2M+18\times0.8}\times d_1=\frac{0.1\times0.8d_1\times1000}{0.1M+0.9\times18}$

$\Rightarrow$ $\frac{5}{0.2M+18\times0.8}=\frac{4}{0.1M+0.9\times18}$

M = 78

20. (2)

f and g are correct

21. (6)

Planar molecules

ClF_3, XeF_2, XeF_4, H_2O, H_2S, OCl_2

22. (3)

Ca, CaH_2 and Ba produce H_2 gas with cold water.

23. (3)

$$H-\underset{\underset{O}{\|}}{C}-OH \xrightarrow[HVZ]{Cl_2/P_4} NO\ \alpha-H, \text{no reaction}$$

$$CH_3-\underset{\underset{O}{\|}}{C}-OH \xrightarrow[HVZ]{Cl_2/P_4} ClCH_2-\underset{\underset{O}{\|}}{C}-OH$$

$$CH_3-\underset{\underset{O}{\|}}{C}-OH \xrightarrow[HVZ]{Cl_2/P_4}$$

$$CH_3-\overset{*}{C}HCl-\underset{\underset{O}{\|}}{C}-OH$$

24. (5)

Chlorine bleaching is based on oxidation, whereas sulphur is based on reduction. Ethanol is less polar as compared to water that's why I_2 is more soluble in it.

As I^- forms complex with I_2 (i.e. I_3^-) which makes I_2 soluble in water. That's why continuous consumption of I^- causes decrease in solubility of I_2.

Alkali metals are at peak, alkaline earth metals are at descending part and halogens are at ascending part of Lother Meyer's Curve.

25. (A, B, C)

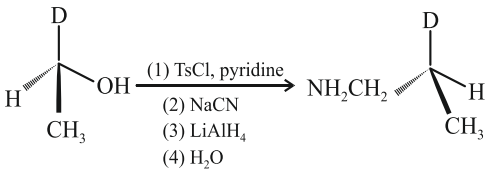
$Al_4C_3 + H_2O \rightarrow Al(OH)_3 + CH_4$

$CaC_2 + H_2O \rightarrow Ca(OH)_2 + C_2H_2$

26. (A, B, C)

(1) De-activated Aromatic compounds do not undergo diazonium coupling.

(2) $H_2N-C_6H_4-OH \xrightarrow{Ac_2O} AcNH-C_6H_4-OH$

(3) 

27. (A, C, D)

$[Cr(H_2O)_6]^{+3}$

$t_2^3e_g^0$ C.F.S.E = –1.2 Δ_0

$[Fe(CN)_6]^{3-}$

$t_2^5e_g^0$ C.F.S.E = –2.0 Δ_0

Ni^{2+} weak field or strong field

C.F.S.E = –1.2 Δ_0

$[Ni(NH_3)_6]^{2+}$

$t_{2g}^6eg^2$ C.F.S.E = –1.2 Δ_0

$\left[Ni\left(H_2O\right)_6\right]^{2+}$

$t_{2g}^6eg^2$

C.F.S.E = –1.2Δ_0

28. (A, B, C, D)

Based on theoretical concepts regarding ionic crystals

29. (B, C)

Anhydrous $FeCl_3$ cannot be obtained by heating of hydrated $FeCl_3$ due to hydrolysis of Fe^{3+} ions.

$2FeCl_3.6H_2O \xrightarrow{\Delta} Fe_2O_3 + 6HCl + 9H_2O$

$FeCl_3.6H_2O + 6SOCl_2$

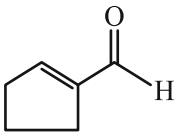
$\rightarrow \underset{\text{Anhydrous}}{FeCl_3} + 6SO_2 + 12HCl$

30. (B, C, D)

Neoprene is a polymer of chloroprene

31. (19.00)

A = OHC – $(CH_2)_4$ – CHO

B = 

MW of B = 96 g mol^{-1}

32. (1.50)

$6HCHO + 4NH_3 \rightarrow \underset{\text{Urotropine}}{\left(CH_2\right)_6 N_4} + 6H_2O$

Number of carbon atom = x = 6

Number of nitrogen atom = y = 4

$\therefore \frac{x}{y} = \frac{6}{4} = 1.5$

33. (6.00)

1000 = a + 0.2b

200 = a + 1.2 b

a = 1160

b = –800

dw = –PdV

$\int dw = \left[a\int dV + b\int VdV\right]$

$w = -\left[a(1.2-0.2) + \frac{b}{2}\left[(1.2)^2 - (0.2)^2\right]\right]$

w = –600 KJ

x = 600 KJ

34. (22.40)

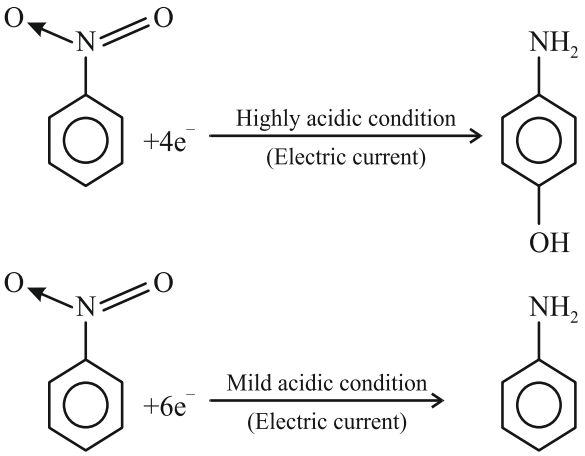
Molar mass of the salt is 322

Moles of salt $= \frac{32.2}{322} = 0.1$

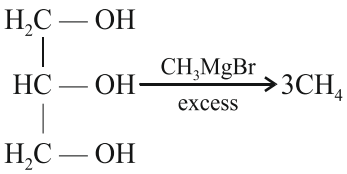
$\therefore$ Moles of oxygen = 14 × 0.1 = 1.4

Mass of oxygen = 1.4 × 16 = 22.4 g

35. (1.50)



36. (2.40)



Weight of CH_4 produced = $\frac{48}{92}$ × 4.6 = 2.4 grams

PART-III (MATHEMATICS)

37. (4)

Let $E = \lim\limits_{n\to\infty}\left({}^{2n}C_n\right)^{\frac{1}{n}}$

$= \lim\limits_{n\to\infty}\left(\dfrac{(n+1)(n+2)(n+3).....(n+n)}{1\cdot 2\cdot 3\cdot .....n}\right)^{1/n}$

$\Rightarrow \quad E = \lim\limits_{n\to\infty}\left(\dfrac{n+1}{1}\cdot\dfrac{n+2}{2}\cdot\dfrac{n+3}{3}.....\dfrac{n+n}{n}\right)^{1/n}$

$\Rightarrow$

$$\ln E = \lim_{n\to\infty}\frac{1}{n}\left[\ln\left(\frac{n+1}{1}\right)+\ln\left(\frac{n+2}{2}\right)+\ln\left(\frac{n+3}{3}\right)+...\right.$$
$$\left.\qquad\qquad .....+\ln\left(\frac{n+n}{n}\right)\right]$$

$= \lim\limits_{n\to\infty}\sum\limits_{r=1}^{n}\dfrac{1}{n}\ln\left(1+\dfrac{n}{r}\right)$

$= \int\limits_0^1 \ln\left(1+\dfrac{1}{x}\right)dx \qquad \left[\dfrac{r}{n}=x,\ \dfrac{1}{n}=dx\right]$

$= \int\limits_0^1 \ln(1+x)\,dx - \int\limits_0^1 \ln x dx$

$= \int\limits_1^2 \ln y\ dy - \int\limits_0^1 \ln x\ dx$

$= \left[y\cdot\ln y - y\right]_1^2 - \left[x\ln x - x\right]_0^1$

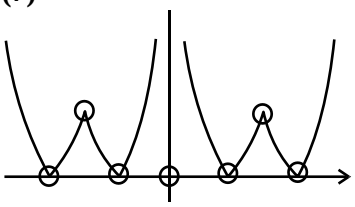
$= \left[(2\ln 2-2)-(0-1)\right]-\left[(-1)-0\right]$

$\Rightarrow$ $\ln E = 2\ln 2 - 2 + 1 + 1$

$\Rightarrow$ $\ln E = 2\ln 2$

$\Rightarrow$ $E = 4$

38. (7)



39. (0)

Either $\sin^{-1}x + \cos^{-1}x + \tan^{-1}x = 0$ or

$\sin^{-1}x = \cos^{-1}x = \tan^{-1}x$

Both of which are not possible $\forall\ x \in R$.

40. (1)

$T_n = \dfrac{H_{n+1}-H_n}{H_nH_{n+1}}$

As $H_{n+1} - H_n = \dfrac{1}{n+1}$

$\Rightarrow T_n = \dfrac{1}{H_n} - \dfrac{1}{H_{n+1}}$

$\Rightarrow T_1 = \dfrac{1}{H_1} - \dfrac{1}{H_2}$

$T_2 = \dfrac{1}{H_2} - \dfrac{1}{H_3}$

$\vdots \quad \vdots \quad \vdots$

$T_n = \dfrac{1}{H_n} - \dfrac{1}{H_{n+1}}$

$T_1 + T_2 + ... + T_{n-1} = \dfrac{1}{H_1} - \dfrac{1}{H_n}$

41. (2)

Since $\alpha^2 - 2\alpha + 3 = 0$ & $\beta^2 - 2\beta + 3 = 0$

$\therefore \quad \alpha^3 - 3\alpha^2 + 5\alpha - 2$

$= \alpha^3 - 2\alpha^2 + 3\alpha - \alpha^2 + 2\alpha - 3 + 1$

$= \alpha(\alpha^2 - 2\alpha + 3) - (\alpha^2 - 2\alpha + 3) + 1 = 1$

And $\beta^3 - \beta^2 + \beta + 5 = \beta^3 - 2\beta^2 + 3\beta + \beta^2 - 2\beta + 5$

$= \beta(\beta^2 - 2\beta + 3) + (\beta^2 - 2\beta + 3) + 2 = 2$

So the ref. equation is

$x^2 - (2 + 1)\,x + 2.1 = 0$

$x^2 - 3x + 2 = 0$

42. (0)

$$\left[\overrightarrow{f(x)}\quad\overrightarrow{g(x)}\quad\overrightarrow{h(x)}\right] = \begin{vmatrix}\cos x & \sin x & \cos 2x\\ \tan x & \sin 3x & \cos 4x\\ \cos 3x & \sin 5x & \cos 6x\end{vmatrix}$$

$\therefore \quad \int\limits_0^{\pi}[f \quad g \quad h](x)dx =$

$\int\limits_0^{\pi}[f \quad g \quad h](\pi - x)dx$

$= -\int\limits_0^{\pi}[f \quad g \quad h](x)\,dx \qquad (\because C_1 \to -C_1)$

$\Rightarrow \quad \int\limits_0^{\pi}[f \quad g \quad h](x)\,dx = 0$

43. (A, C)

$I_{n+1} - I_n = \int\limits_0^{\pi}\dfrac{1-\cos(n+1)x-1+\cos nx}{1-\cos x}dx$

$= \int\limits_0^{\pi}\dfrac{\sin\left(n+\frac{1}{2}\right)x}{\sin\frac{x}{2}}\,dx \qquad ....(i)$

Similarly

$I_n - I_{n-1} = \int\limits_0^{\pi}\dfrac{\sin\left(n-\frac{1}{2}\right)x}{\sin\frac{x}{2}}dx \qquad ....(ii)$

On subtracting (ii) from (i)

$$I_{n+1} + I_{n-1} - 2I_n = \int_0^{\pi} \frac{\sin\left(n+\frac{1}{2}\right)x - \sin\left(n-\frac{1}{2}\right)x}{\sin\frac{x}{2}}\,dx$$

$$= \int_0^{\pi} 2\cos nx\,dx = 0$$

$$\Rightarrow \quad I_{n+1} + I_{n-1} = 2I_n$$

$$I_1 = \int_0^{\pi} 1\,dx = \pi$$

$$I_2 = \int_0^{\pi} \frac{1-\cos 2x}{1-\cos x}\,dx = 2\pi$$

Hence $I_n = n\pi\ ;\ n \geq 1$

44. (B, D)

Let $f(x) = (1+x)^{2000}$

then $f(1) + \omega f(\omega) + \omega^2 f(\omega^2)$

$= 3\left({}^{2000}C_2 + {}^{2000}C_5 + {}^{2000}C_8 + {}^{2000}C_{11} \ldots.\right)$

$\Rightarrow \ {}^{2000}C_2 + {}^{2000}C_5 + {}^{2000}C_8 + \ldots. = \frac{2^{2000}-1}{3}$

$\left\{\text{where } \omega = -\frac{1}{2} + \frac{i\sqrt{3}}{2}\right\}$

45. (C, D)

The perpendicular line to both given line will be

$\frac{x+1}{-2} = \frac{y+2}{0} = \frac{z-3}{1}$

A point can be taken on this line $(-2t-1, -2, t+3)$

So, perpendicular distance is $2\sqrt{3}$ from $x+y+z-3=0$, then $t = -9, 3$

Corresponding points are (17, –2, –6), (–7, –2, 6)

46. (A, C, D)

$$x\frac{dy}{dx} - 2y = 2 - 4\ln x$$

$$\Rightarrow \frac{dy}{dx} - \frac{2}{x}y = \frac{2-4\ln x}{x}$$

I.F. $= e^{-\int \frac{2}{x}dx} = \frac{1}{x^2}$

$$\therefore \ y\frac{1}{x^2} = \int \frac{2-4\ln x}{x^3}\,dx$$

$$\Rightarrow \frac{y}{x^2} = -\frac{1}{x^2} - 4\left(\frac{\ln x x^{-2}}{-2} + \frac{x^{-2}}{-4}\right) + c$$

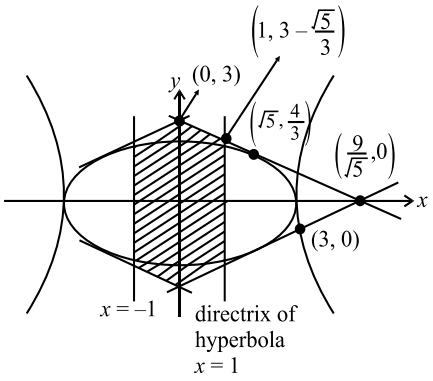
$$\Rightarrow y = cx^2 + 2\ln x$$

$\because$ It passes through (1, 3)

$\therefore \ c = 3$

$\therefore \ y = 3x^2 + 2\ln x$

47. (A, D)



$$\frac{x^2}{9} + \frac{y^2}{4} = 1$$

$$e = \sqrt{1-\frac{4}{9}} = \frac{\sqrt{5}}{3}$$

$$\frac{x^2}{9} - \frac{y^2}{72} = 1$$

$$e = \sqrt{1+\frac{72}{9}} = 3$$

$$\frac{x^2}{9} + \frac{y^2}{4} = 1$$

$$\Rightarrow \quad \frac{x \times \sqrt{5}}{9} + \frac{y \times \frac{4}{3}}{4} = 1$$

$$\Rightarrow \quad \frac{\sqrt{5}x}{9} + \frac{y}{3} = 1$$

If $x = 1$

$$y = 3\left(1 - \frac{\sqrt{5}}{9}\right)$$

$$\therefore \quad \left(1, 3-\frac{\sqrt{5}}{3}\right)$$

Area $= 2 \times \left(\frac{1}{2} \times 2 \times \frac{\sqrt{5}}{3}\right) + 2 \times 6\left(1 - \frac{\sqrt{5}}{9}\right)$

$$= \frac{2\sqrt{5}}{1} + 12 - \frac{4\sqrt{5}}{3}$$

$$= 12 - \frac{2\sqrt{5}}{3} = 10.51$$

$\therefore \quad [A] = 10$

48. (A, C)

$(MN)^2 = 3MN$

$\Rightarrow NMNMNM = 3NMNM$

$\Rightarrow (NM)^3 = 3(NM)^2 \Rightarrow (NM) = 3I$

$$P=\frac{1}{3}I \text{ so, } P+P^2+....=\left(\frac{1}{3}+\frac{1}{3^2}+....\right)I=\frac{1}{2}I$$

49. (6.00)

No. of reflexive relations = $2^{16-4}=2^{12}$

50. (0.50)

$$S=\tan^{-1}\left(\frac{1}{2\sqrt{2}}\right)+\tan^{-1}\left(\frac{1}{4\sqrt{2}}\right)+\tan^{-1}\left(\frac{1}{7\sqrt{2}}\right)+.....$$

$$\tan^{-1}\left(\frac{\left(\frac{2}{\sqrt{2}}-\frac{1}{\sqrt{2}}\right)}{1+\frac{2}{\sqrt{2}}\cdot\frac{1}{\sqrt{2}}}\right) \quad +\tan^{-1}\left[\frac{\frac{3}{\sqrt{2}}-\frac{2}{\sqrt{2}}}{1+\frac{3}{\sqrt{2}}\cdot\frac{2}{\sqrt{2}}}\right]+$$

.....∞ terms

$$=\frac{\pi}{2}-\tan^{-1}\frac{1}{\sqrt{2}}=\frac{\pi}{2}-\cot^{-1}\sqrt{2}\ =\tan^{-1}\left(\sqrt{2}\right)$$

51. (18.00)

$$\frac{\sum x_i}{N}=17$$

$$\therefore \quad \frac{\sum x_i+\frac{2.N}{2}}{N}=18$$

52. (1.00)

$$\Rightarrow \quad \sin x>\frac{1}{2}$$

$$\therefore \quad x\in\left(\frac{\pi}{6},\frac{5\pi}{6}\right)$$

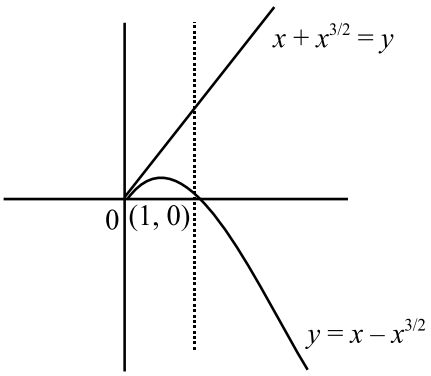
53. (0.80)

$(y-x)^2=x^3, x\geq 0$

$\Rightarrow y=x\pm x^{3/2}$

Required area

$$=\int_0^1\left(\left(x+x^{3/2}\right)-\left(x-x^{3/2}\right)\right)dx=\frac{4}{5} \text{ sq. unit}$$



54. (0.20)

$f(g(x)) = x$

$[f'(g(x))]\cdot g'(x) = 1$

$$g'(x)=\frac{1}{f'(g(x))}$$

$$g'\left(-\frac{7}{6}\right)=\frac{1}{f'(1)}$$

$$f(1)=-\frac{7}{6};\ f'(1)=5$$

$$\Rightarrow g\left(-\frac{7}{6}\right)=\frac{1}{5}=0.2$$

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